

The Garden of Eden--plant based diets, the genetic drive to conserve cholesterol and its implications for heart disease in the 21st century.

It is likely that plant food consumption throughout much of human evolution shaped the dietary requirements of contemporary humans. Diets would have been high in dietary fiber, vegetable protein, plant sterols and associated phytochemicals, and low in saturated and trans-fatty acids and other substrates for cholesterol biosynthesis. To meet the body's needs for cholesterol, we believe genetic differences and polymorphisms were conserved by evolution, which tended to raise serum cholesterol levels. As a result modern man, with a radically different diet and lifestyle, especially in middle age, is now recommended to take medications to lower cholesterol and reduce the risk of cardiovascular disease. Experimental introduction of high intakes of viscous fibers, vegetable proteins and plant sterols in the form of a possible Myocene diet of leafy vegetables, fruit and nuts, lowered serum LDL-cholesterol in healthy volunteers by over 30%, equivalent to first generation statins, the standard cholesterol-lowering medications. Furthermore, supplementation of a modern therapeutic diet in hyperlipidemic subjects with the same components taken as oat, barley and psyllium for viscous fibers, soy and almonds for vegetable proteins and plant sterol-enriched margarine produced similar reductions in LDL-cholesterol as the Myocene-like diet and reduced the majority of subjects' blood lipids concentrations into the normal range. We conclude that reintroduction of plant food components, which would have been present in large quantities in the plant based diets eaten throughout most of human evolution into modern diets can correct the lipid abnormalities associated with contemporary eating patterns and reduce the need for pharmacological interventions.